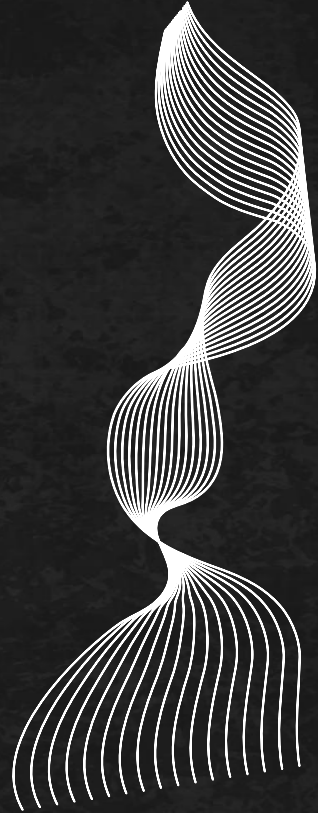


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# Industrial Manufacturer Network design Final



By Tanner, Rin, Yu-An (Annie), Caiqing, Adarsh



# Meet The Team



**Network Architect**

Yu-An (Annie)



**Industrial  
Cybersecurity Expert**

Tanner



**Network Analyst**

Rin



**Cyber Risk Specialist**

Adarsh



**Network Engineer**

Caiqing



# Layer 1 Recommendations



- Replace unmanaged switches with managed, rack-mounted switches
- Use locked network cabinets and install key card readers for network closets (restricted to network admins)
- Use color-coded, shielded cabling with labeled ports for auditability
- Segment departments physically where possible
- Enable port-based authentication with 802.1x

# Layer 2 Recommendations



- Implement VLAN Tagging (802.1q) and configure trunk ports to support inter-vlan routing
  - With 802.1x Authentication using FreeRADIUS server
- Port security by limiting MAC Addresses and disabling unused ethernet ports
  - One to two Mac addresses per port
  - Auto - shutdown when rogue device is detected
- Two core switches stacked



# VLANs



|    |              |     |                                              |
|----|--------------|-----|----------------------------------------------|
| 10 | IT (Systems) | 70  | HR                                           |
| 20 | Accounting   | 80  | Guest Wifi                                   |
| 30 | Marketing    | 90  | Network Management                           |
| 40 | Sales        | 110 | E-commerce DB                                |
| 50 | Executive    | 120 | Internal DNS/DHCP/DC,<br>File, Wazuh, Zabbix |
| 60 | R&D          |     |                                              |



# Layer 3 Recommendations

## Overall Security Strategy



Move internal systems to RFC1918 address space



Enforce NAT boundaries



Implement segmentation by business function



Strict Firewall Controls regulating inbound and outbound traffic

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# Layer 3 Recommendations



## Network Segmentation Design

- **Multihomed Firewall Deployment**
  - Firewall interfaces directly connect to each zone
- **Logical Zones Created**
  - Internal Trusted Network: Workstations, internal servers, network management
  - DMZ (Demilitarized Zone): Public-facing servers (Web, Public DNS, Mail)
  - Internet Edge: External untrusted traffic

# Layer 3 Recommendations



## Critical Services Protection

- **External DNS (testcorp.com)** for public web servers
  - A Records:
    - www.testcorp.com → 204.130.145.10 (Web Server)
    - mail.testcorp.com → 204.130.145.20 (Mail Server)
    - dns.testcorp.com → 204.130.145.30 (Public DNS Server)
  - MX Record:
    - testcorp.com → mail.testcorp.com (priority 10)
- **Internal DNS (testcorp.local)**
  - Internal domain name resolution
  - Forwards external queries securely to external DNS



# Layer 3 Recommendations



## **Access Control**

- ACLs for inter-VLAN routing
- Organizational Units(OUs) created and Group Policy Objects(GPOs) assigned for server hardening

Note: Please see the following slides for all configuration details (ex. NAT, FW rules, DHCP scope, OUs, GPOs etc.)

# NAT Table



| Direction    | Source IP Range | Destination (Public IP) | Translated To  | Method     | Use Case                               |
|--------------|-----------------|-------------------------|----------------|------------|----------------------------------------|
| Outbound     | 192.168.0.0/16  | Any                     | 204.130.145.1  | PAT        | Internal devices >> Internet           |
| Inbound HTTP | Any             | 204.130.145.10          | 192.168.100.10 | Static NAT | Web server (E-commerce) (port 80, 443) |
| Inbound SMTP | Any             | 204.130.145.20          | 192.168.100.20 | Static NAT | Mail server (port 25, 465)             |
| Inbound DNS  | Any             | 204.130.145.30          | 192.168.100.30 | Static NAT | Public DNS (port 53)                   |

# Next-gen Firewall Rule - Inbound



| # | Source                                                                                                                          | Destination      | Protocol/Port | Action     | Purpose                                            |
|---|---------------------------------------------------------------------------------------------------------------------------------|------------------|---------------|------------|----------------------------------------------------|
| 1 | RFC1918 (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16) and Bogon IPs (0.0.0.0/8, 127.0.0.0/8, 169.254.0.0/16, 100.64.0.0/10, etc.) | Any              | Any           | Deny       | Block invalid source IPs (anti-spoofing)           |
| 2 | Any                                                                                                                             | 204.130.145.10   | TCP 80, 443   | Allow      | Web server access                                  |
| 3 | Any                                                                                                                             | 204.130.145.20   | TCP 25, 465   | Allow      | Mail server access                                 |
| 4 | Any                                                                                                                             | 204.130.145.30   | UDP 53        | Allow      | DNS, resolving <i>testcorp.com</i>                 |
| 5 | 192.168.100.10                                                                                                                  | 192.168.110.10   | TCP 1433      | Allow      | Web Server access to E-commerce DB (Suppose MSSQL) |
| 6 | 192.168.100.0/24                                                                                                                | 192.168.110.0/24 | Any           | Deny       | Block all other DMZ to DB VLAN traffic             |
| 7 | Any                                                                                                                             | 192.168.0.0/16   | Any           | Deny       | Protect Internal network                           |
| 8 | Any                                                                                                                             | Any              | IDS/IPS       | Drop/Alert | Block attacks                                      |
| 9 | Any                                                                                                                             | Any              | Any           | Deny       | Default rule                                       |

# Next-gen Firewall Rule - Outbound



| # | Source                                | Destination                 | Protocol/Port    | Action | Purpose                                         |
|---|---------------------------------------|-----------------------------|------------------|--------|-------------------------------------------------|
| 1 | 192.168.110.0/24 (E-commerce DB VLAN) | Any                         | Any              | Deny   | Block DB server outbound Internet access        |
| 2 | 192.168.80.0/24 (Guest Wi-Fi)         | 192.168.0.0/16              | Any              | Deny   | Guest isolation from Internal LAN               |
| 3 | 192.168.0.0/16                        | Any                         | TCP 80, 443      | Allow  | Internal web browsing (HTTP/HTTPS)              |
| 4 | 192.168.0.0/16                        | Any                         | UDP 123          | Allow  | Internal NTP (time sync)                        |
| 5 | 192.168.0.0/16                        | Any                         | TCP 25, 465, 587 | Allow  | Internal email outbound                         |
| 6 | 192.168.120.10 (Internal DNS)         | 192.168.100.30 (Public DNS) | UDP/TCP 53       | Allow  | Internal DNS forwarding to Public DNS           |
| 7 | 192.168.0.0/16                        | Any                         | UDP/TCP 53       | Deny   | Block Internal DNS queries directly to Internet |
| 8 | Any                                   | Any                         | Any              | Deny   | Default deny for all other outbound traffic     |



# DHCP Scope



| Scope Name        | VLAN | Subnet          | IP Range (Pool)      | Default Gateway | DNS Server     | Notes                                                                           |
|-------------------|------|-----------------|----------------------|-----------------|----------------|---------------------------------------------------------------------------------|
| IT (Systems)      | 10   | 192.168.10.0/24 | 192.168.10.10 – .239 | 192.168.10.1    | 192.168.120.10 | IT/Systems Admins; <b>.240–.250 reserved for Privileged Admins (static IPs)</b> |
| Accounting Scope  | 20   | 192.168.20.0/24 | 192.168.20.10 – .200 | 192.168.20.1    | 192.168.120.10 | For Accounting devices                                                          |
| Marketing Scope   | 30   | 192.168.30.0/24 | 192.168.30.10 – .200 | 192.168.30.1    | 192.168.120.10 | For Marketing department                                                        |
| Sales Scope       | 40   | 192.168.40.0/24 | 192.168.40.10 – .200 | 192.168.40.1    | 192.168.120.10 | For Sales devices                                                               |
| Executive Scope   | 50   | 192.168.50.0/24 | 192.168.50.10 – .200 | 192.168.50.1    | 192.168.120.10 | For Executive staff                                                             |
| R&D Scope         | 60   | 192.168.60.0/24 | 192.168.60.10 – .200 | 192.168.60.1    | 192.168.120.10 | For Engineering/R&D devices                                                     |
| HR                | 70   | 192.168.70.0/24 | 192.168.70.10 – .200 | 192.168.70.1    | 192.168.120.10 | For HR devices                                                                  |
| Guest Wi-Fi Scope | 80   | 192.168.80.0/24 | 192.168.80.10 – .200 | 192.168.80.1    | 192.168.120.10 | For Guest WiFi                                                                  |

Note: VLANs 90, 110, and 120 use static IP addressing.

VLAN 90 ensures secure management of network devices, while VLANs 110 and 120 provide stable and reliable access to critical database and internal server resources.

# ACLs

| VLAN 10 - IT (Systems) |                   |                                  |                   |                                                                  |
|------------------------|-------------------|----------------------------------|-------------------|------------------------------------------------------------------|
| #                      | Source            | Destination                      | Protocol/Port     | Action                                                           |
| 1                      | 192.168.10.240/28 | 192.168.90.0/24                  | TCP 22, 443, SNMP | Allow (Only Privileged Admins are allowed in Network Management) |
| 2                      | 192.168.10.0/24   | 192.168.120.10                   | UDP 53, 67        | Allow (DNS/DHCP/DC access)                                       |
| 3                      | 192.168.10.0/24   | 192.168.120.20                   | TCP 445           | Allow (File server access)                                       |
| 4                      | 192.168.10.0/24   | 192.168.120.30 (osTicket server) | TCP 80, 443       | Allow (IT Helpdesk management)                                   |
| 5                      | 192.168.10.0/24   | 192.168.120.40 (Wazuh server)    | TCP 443           | Allow (SIEM dashboard access)                                    |
| 6                      | 192.168.10.0/24   | 192.168.120.50 (Zabbix server)   | TCP 443           | Allow (Network monitoring access)                                |
| 7                      | 192.168.10.0/24   | Internet                         | TCP 80, 443       | Allow (Web browsing)                                             |
| 8                      | 192.168.10.0/24   | Any                              | Any               | Deny                                                             |
|                        |                   |                                  |                   |                                                                  |
| VLAN 20 - Accounting   |                   |                                  |                   |                                                                  |
| #                      | Source            | Destination                      | Protocol/Port     | Action                                                           |
| 1                      | 192.168.20.0/24   | 192.168.120.10                   | UDP 53, 67        | Allow                                                            |
| 2                      | 192.168.20.0/24   | 192.168.120.20                   | TCP 445           | Allow                                                            |
| 3                      | 192.168.20.0/24   | 192.168.120.30                   | TCP 80, 443       | Allow                                                            |
| 4                      | 192.168.20.0/24   | Internet                         | TCP 80, 443       | Allow                                                            |
| 5                      | 192.168.20.0/24   | Any                              | Any               | Deny                                                             |
|                        |                   |                                  |                   |                                                                  |
| VLAN 30 - Marketing    |                   |                                  |                   |                                                                  |
| #                      | Source            | Destination                      | Protocol/Port     | Action                                                           |
| 1                      | 192.168.30.0/24   | 192.168.120.10                   | UDP 53, 67        | Allow                                                            |
| 2                      | 192.168.30.0/24   | 192.168.120.20                   | TCP 445           | Allow                                                            |
| 3                      | 192.168.30.0/24   | 192.168.120.30                   | TCP 80, 443       | Allow                                                            |
| 4                      | 192.168.30.0/24   | Internet                         | TCP 80, 443       | Allow                                                            |
| 5                      | 192.168.30.0/24   | Any                              | Any               | Deny                                                             |

# ACLs (cont.)

## VLAN 40 – Sales

| # | Source          | Destination    | Protocol/Port | Action |
|---|-----------------|----------------|---------------|--------|
| 1 | 192.168.40.0/24 | 192.168.120.10 | UDP 53, 67    | Allow  |
| 2 | 192.168.40.0/24 | 192.168.120.20 | TCP 445       | Allow  |
| 3 | 192.168.40.0/24 | 192.168.120.30 | TCP 80, 443   | Allow  |
| 4 | 192.168.40.0/24 | Internet       | TCP 80, 443   | Allow  |
| 5 | 192.168.40.0/24 | Any            | Any           | Deny   |

## VLAN 50 – Executive

| # | Source          | Destination    | Protocol/Port | Action |
|---|-----------------|----------------|---------------|--------|
| 1 | 192.168.50.0/24 | 192.168.120.10 | UDP 53, 67    | Allow  |
| 2 | 192.168.50.0/24 | 192.168.120.20 | TCP 445       | Allow  |
| 3 | 192.168.50.0/24 | 192.168.120.30 | TCP 80, 443   | Allow  |
| 4 | 192.168.50.0/24 | Internet       | TCP 80, 443   | Allow  |
| 5 | 192.168.50.0/24 | Any            | Any           | Deny   |

## VLAN 60 – R&D

| # | Source          | Destination    | Protocol/Port | Action |
|---|-----------------|----------------|---------------|--------|
| 1 | 192.168.60.0/24 | 192.168.120.10 | UDP 53, 67    | Allow  |
| 2 | 192.168.60.0/24 | 192.168.120.20 | TCP 445       | Allow  |
| 3 | 192.168.60.0/24 | 192.168.120.30 | TCP 80, 443   | Allow  |
| 4 | 192.168.60.0/24 | Internet       | TCP 80, 443   | Allow  |
| 5 | 192.168.60.0/24 | Any            | Any           | Deny   |

## VLAN 70 – HR

| # | Source          | Destination    | Protocol/Port | Action |
|---|-----------------|----------------|---------------|--------|
| 1 | 192.168.70.0/24 | 192.168.120.10 | UDP 53, 67    | Allow  |
| 2 | 192.168.70.0/24 | 192.168.120.20 | TCP 445       | Allow  |
| 3 | 192.168.70.0/24 | 192.168.120.30 | TCP 80, 443   | Allow  |
| 4 | 192.168.70.0/24 | Internet       | TCP 80, 443   | Allow  |
| 5 | 192.168.70.0/24 | Any            | Any           | Deny   |

# ACLs (cont.)

| <b>VLAN 80 – Guest Wi-Fi</b>        |                                   |                                      |                   |                                                  |
|-------------------------------------|-----------------------------------|--------------------------------------|-------------------|--------------------------------------------------|
| #                                   | Source                            | Destination                          | Protocol/Port     | Action                                           |
| 1                                   | 192.168.80.0/24                   | Internet                             | TCP 80, 443       | Allow                                            |
| 2                                   | 192.168.80.0/24                   | 192.168.0.0/16                       | Any               | Deny                                             |
| <b>VLAN 90 – Network Management</b> |                                   |                                      |                   |                                                  |
| #                                   | Source                            | Destination                          | Protocol/Port     | Action                                           |
| 1                                   | 192.168.10.240/28                 | 192.168.90.0/24                      | TCP 22, 443, SNMP | Allow (Only Privileged IT IPs allowed to manage) |
| 2                                   | 192.168.90.0/24                   | Network Devices (Firewall, Switches) | TCP 22, 443, SNMP | Allow (Internal device-to-device management)     |
| 3                                   | 192.168.90.0/24                   | Any                                  | Any               | Deny                                             |
| <b>VLAN 110 – E-commerce DB</b>     |                                   |                                      |                   |                                                  |
| #                                   | Source                            | Destination                          | Protocol/Port     | Action                                           |
| 1                                   | Any Internal VLAN                 | 192.168.110.0/24                     | Any               | Deny                                             |
| <b>VLAN 120 – Internal Servers</b>  |                                   |                                      |                   |                                                  |
| #                                   | Source                            | Destination                          | Protocol/Port     | Action                                           |
| 1                                   | 192.168.10.0/24 – 192.168.70.0/24 | 192.168.120.10 (DNS/DHCP/DC)         | UDP 53, 67        | Allow                                            |
| 2                                   | 192.168.10.0/24 – 192.168.70.0/24 | 192.168.120.20 (File server)         | TCP 445           | Allow                                            |
| 3                                   | 192.168.10.0/24 – 192.168.70.0/24 | 192.168.120.30 (osTicket server)     | TCP 80, 443       | Allow                                            |
| 4                                   | 192.168.10.0/24                   | 192.168.120.40 (Wazuh server)        | TCP 443           | Allow                                            |
| 5                                   | 192.168.10.0/24                   | 192.168.120.50 (Zabbix server)       | TCP 443           | Allow                                            |
| 6                                   | 192.168.80.0/24                   | 192.168.120.0/24                     | Any               | Deny                                             |
| 7                                   | 192.168.120.0/24                  | Any                                  | Any               | Deny                                             |

# OUs



| OU           | Purpose                                                                      |
|--------------|------------------------------------------------------------------------------|
| IT (Systems) | Privileged Admins and Helpdesk users                                         |
| Accounting   | Accounting users                                                             |
| Marketing    | Marketing users                                                              |
| Sales        | Sales users                                                                  |
| Executives   | Executive users                                                              |
| R&D          | Research and Development users                                               |
| HR           | Human Resources users                                                        |
| Servers      | All Internal Servers (DNS/DHCP/DC, File, Wazuh, Zabbix, E-commerce Database) |
| Guests       | Guest users using WiFi                                                       |



# GPOs

| GPO Name                    | Linked OU                                              | Settings                                                                                                                                                                                                                                                                                                                                                              | Purpose                                                                                       |
|-----------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Default Domain Policy       | Entire Domain                                          | <ul style="list-style-type: none"><li>- Minimum password length (12 characters)</li><li>- Password complexity required</li><li>- Password expiration (90 days)</li><li>- Account lockout threshold (5 failed attempts)</li><li>- Lockout duration (15 minutes)</li><li>- Reset lockout counter after 15 minutes</li><li>- Kerberos ticket lifetime settings</li></ul> | Enforce basic security settings across the entire domain (users and computers).               |
| IT Admins Security Policy   | OU: IT (Systems)                                       | <ul style="list-style-type: none"><li>- Allow local admin rights</li><li>- Allow installation of software</li><li>- Enforce Windows Firewall ON</li><li>- Require antivirus</li><li>- Enable management tools access (Remote Server Admin Tools)</li></ul>                                                                                                            | Allow IT admins full workstation/server control while maintaining basic endpoint protections. |
| Workstation Security Policy | OUs: Accounting, Marketing, Sales, R&D, HR, Executives | <ul style="list-style-type: none"><li>- Enforce Windows Firewall ON</li><li>- Require antivirus</li><li>- Disable USB storage</li><li>- Remove local admin rights</li><li>- Disable unnecessary services (Bluetooth, etc.)</li><li>- Restrict Control Panel access</li></ul>                                                                                          | Harden normal employee computers, prevent lateral movement, ensure endpoint protection.       |
| Server Hardening Policy     | OU: Servers                                            | <ul style="list-style-type: none"><li>- Enforce Windows Firewall ON</li><li>- Require antivirus</li><li>- Deny interactive logon for normal users</li><li>- Only IT Admins can RDP into servers</li><li>- Disable guest account</li><li>- Require automatic updates enabled</li></ul>                                                                                 | Secure internal DNS, DHCP, File, and DB servers to reduce attack surface.                     |
| Guest Wi-Fi Device Policy   | OU: Guests                                             | <ul style="list-style-type: none"><li>- Block drive mappings</li><li>- Block access to internal servers</li><li>- Allow DNS, NTP, Internet only</li></ul>                                                                                                                                                                                                             | Protect internal resources from Guest Wi-Fi devices even if domain joined.                    |

# Industrial Manufacturing Additional Considerations



- Isolate OT systems using dedicated VLANs and firewalled Layer 3 routing
- Separate IT and OT networks
- Deploy intrusion detection for ICS protocols (e.g., Modbus, DNP3)
- Disable auto-routing and USB device access on ICS assets
- Require jump servers for remote access to ICS environments

# Policies

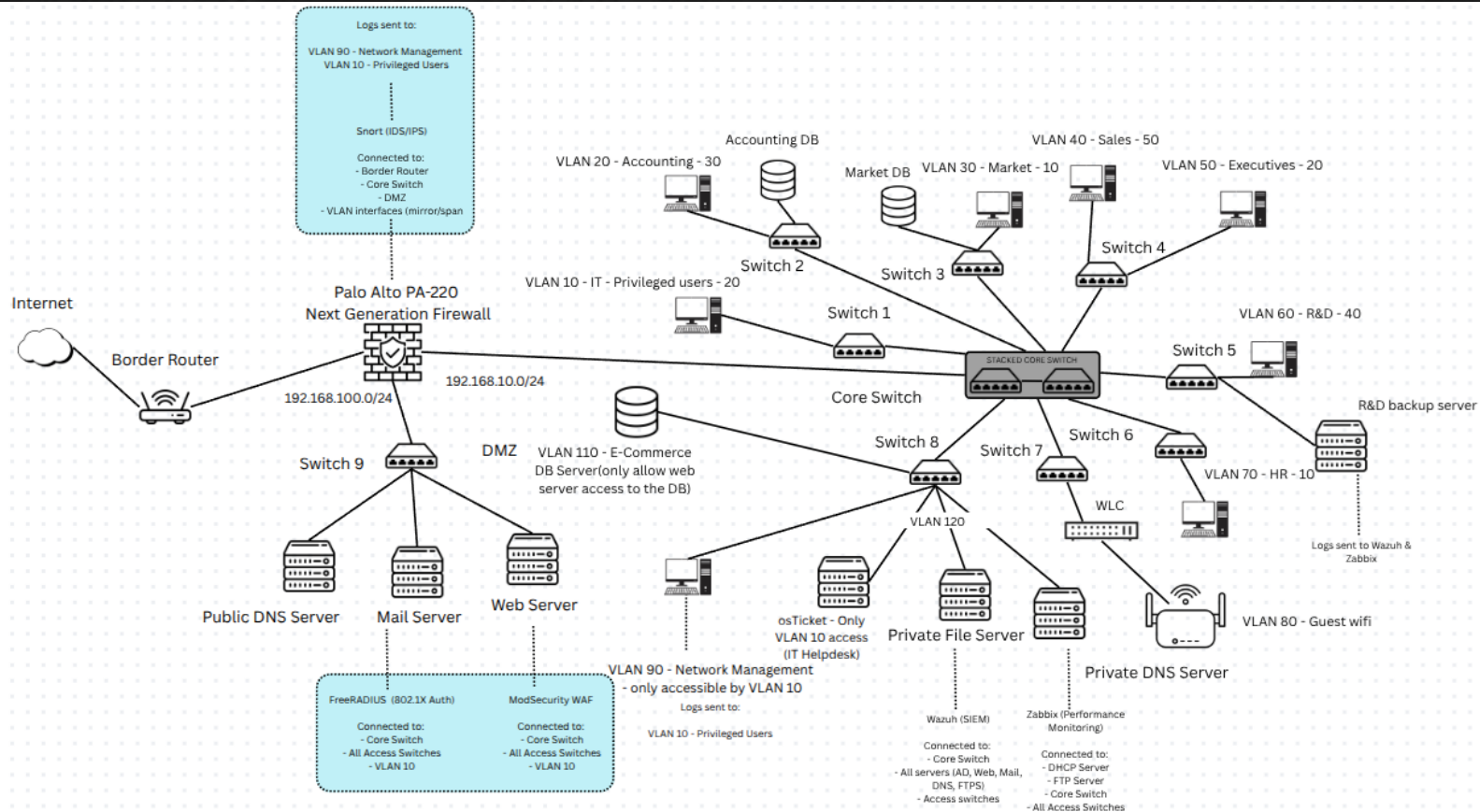
| Category            | Policy Name                          | Policy Description                                                                                                                                                             |
|---------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer 1 – Physical  | Physical Access Control Policy       | Network equipment must be housed in secure rooms, accessible only by authorized IT personnel with ID verification and access log auditing. Ports must be disabled when unused. |
|                     | Cabling and Rack Management Policy   | Cabling must follow structured guidelines, clearly labeled by department. Racks must have power redundancy (UPS) and proper cable management.                                  |
|                     | Device Inventory and Port Mapping    | Switches/routers must maintain port maps with MAC addresses and department mappings. Devices should be tracked in real-time using SNMP-based monitoring.                       |
| Layer 2 – Data Link | Inter-VLAN Traffic Control Policy    | Inter-VLAN traffic is allowed only based on business need and enforced through ACLs.                                                                                           |
|                     | Port Security Policy                 | Switch ports must allow a max of 2 MAC addresses per port. Violations trigger alerts and optional port shutdown.                                                               |
|                     | Enterprise Wireless Policy           | Wireless access must use WPA3-Enterprise with 802.1X and RADIUS authentication. Wireless SSIDs must be segmented and mapped to VLAN 80 for guest use.                          |
|                     | BYOD Policy – Personal Router Ban    | Personal wireless routers are prohibited. Unauthorized devices detected are blocked and logged, with violations resulting in disciplinary action.                              |
| Layer 3 – Network   | Routing and ACLs Policy              | Inter-VLAN routing handled by Layer 3 switches/NGFW. ACL changes reviewed monthly and NAT mappings documented and reviewed quarterly.                                          |
|                     | Remote Access Policy                 | No direct remote access to internal resources. VPN services are disabled unless project-specific and justified.                                                                |
| Software & Patch    | Software and Patch Management Policy | Critical patches applied within 24 hours, high severity in 3 days, routine in 7 days. Automated monthly vulnerability scanning.                                                |
|                     | Application Whitelisting Policy      | Only approved applications allowed. All installations must be logged and reviewed.                                                                                             |
| Employee Security   | Security Awareness Training Policy   | Mandatory training on phishing, password hygiene, data classification, and incident reporting upon hiring, annually, and after incidents.                                      |
|                     | Acceptable Use Policy (AUP)          | Employees must not install unauthorized software, use personal devices, or bypass security controls. Corporate asset use must align with company guidelines.                   |
|                     | Incident Response & Reporting Policy | Security incidents must be reported to IT within 30 minutes. IT responds and documents incidents per the IR plan.                                                              |
| Department-Specific | Department-Specific Restrictions     | R&D/Executive VLANs are isolated from marketing/sales. Sensitive databases require MFA and logging for access.                                                                 |



# URL FILTERING

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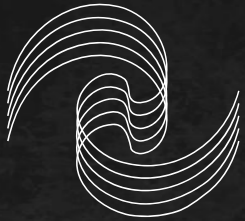
# Network infrastructure





# VLAN Configuration

| VLAN | Department                                | Subnet           | Gateway IP    |  |
|------|-------------------------------------------|------------------|---------------|--|
| 10   | IT (Systems)                              | 192.168.10.0/24  | 192.168.10.1  |  |
| 20   | Accounting                                | 192.168.20.0/24  | 192.168.20.1  |  |
| 30   | Marketing                                 | 192.168.30.0/24  | 192.168.30.1  |  |
| 40   | Sales                                     | 192.168.40.0/24  | 192.168.40.1  |  |
| 50   | Executive                                 | 192.168.50.0/24  | 192.168.50.1  |  |
| 60   | R&D                                       | 192.168.60.0/24  | 192.168.60.1  |  |
| 70   | HR                                        | 192.168.70.0/24  | 192.168.70.1  |  |
| 80   | Guest Wi-Fi                               | 192.168.80.0/24  | 192.168.80.1  |  |
| 90   | Network Management                        | 192.168.90.0/24  | 192.168.90.1  |  |
| 110  | E-commerce DB                             | 192.168.110.0/24 | 192.168.110.1 |  |
| 120  | Internal DNS/DHCP/DC, File, Wazuh, Zabbix | 192.168.120.0/24 | 192.168.120.1 |  |



**\$43,349.98**



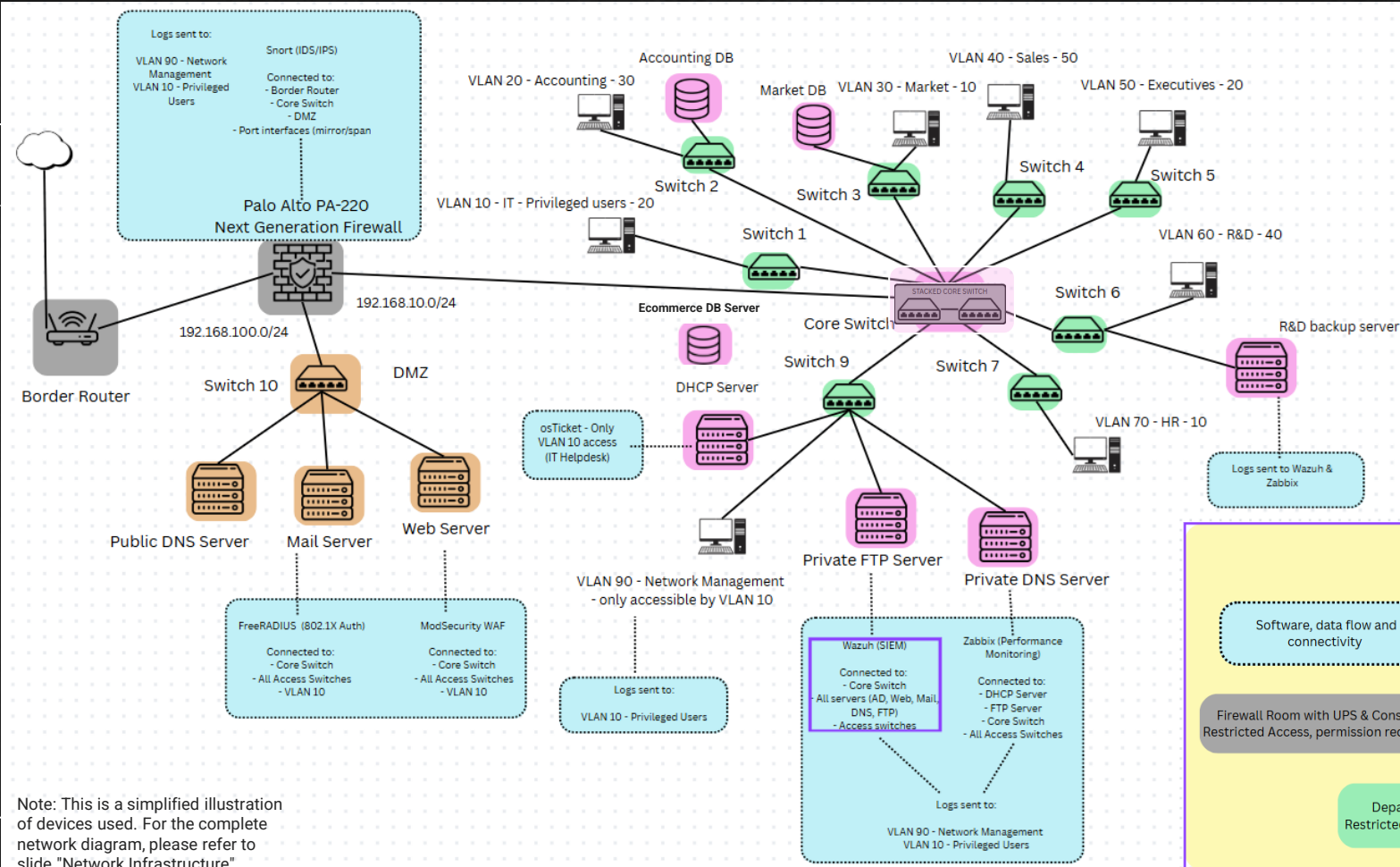
Cost Estimates for Secure Network Overhaul

# Cost Estimates

| Item                                     | Model / Description                                                                                                   | Qty              | Unit Price | Total Price | Physical Location                                | Cabling Type                                | Notes                                                     |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------|------------|-------------|--------------------------------------------------|---------------------------------------------|-----------------------------------------------------------|
| Core Switch                              | Cisco Business 350 Series 24-Port Gigabit Managed Switch (CBS350-24T-4G)                                              | 2                | \$500      | \$1,000     | Main Server Room (Central Rack)                  | Cat6a for access switches; Fiber (optional) | Layer 3 switch with VLAN support                          |
| Access Switches                          | Cisco Business 250 Series 48-Port Gigabit Smart Switch (CBS250-48T-4G)                                                | 5 (+ 4 existing) | \$400      | \$2,000     | Each department's network closet                 | Cat6a to endpoints                          | Provides PoE+ for devices like IP phones and cameras      |
| Next-Gen Firewall                        | Palo Alto PA-220                                                                                                      | 1                | \$1,364.98 | \$1,364.98  | Between ISP modem and core switch                | Cat6a                                       | Advanced threat prevention features                       |
| Access Control System                    | HID Global Edge EVO Solo ESHR40-K                                                                                     | 3                | \$1,200    | \$3,600     | Entrances to server room, DMZ, and IT admin area | Cat6a for network connectivity              | Standalone IP-based access control                        |
| Structured Cabling                       | Cat6a Plenum-Rated Cable (1000ft)                                                                                     | 10               | \$200      | \$2,000     | Throughout the facility                          | N/A                                         | Supports up to 10Gbps over 100 meters                     |
| Patch Panels and Accessories             | 24-Port Cat6a Patch Panel                                                                                             | 10               | \$100      | \$1,000     | Network closets and server room                  | N/A                                         | Organizes and manages cable connections                   |
| Rack Mounts                              | 42U Server Rack Cabinet                                                                                               | 2                | \$1,000    | \$2,000     | Server room and main network closet              | N/A                                         | Houses core networking equipment                          |
| Uninterruptible Power Supply (UPS)       | APC Smart-UPS 1500VA LCD 120V                                                                                         | 3                | \$599      | \$1,797     | Server room and main network closet              | N/A                                         | Provides backup power during outages                      |
| Managed Security Service Provider (MSSP) | GhostWatch MSSP Subscription                                                                                          | 12 months        | \$1,999    | \$23,988    | N/A                                              | N/A                                         | 24/7 monitoring and incident response                     |
| Warranty and Support                     | Cisco Smart Net Total Care (3-Year)                                                                                   | 1                | \$1,000    | \$1,000     | N/A                                              | N/A                                         | Covers all Cisco networking equipment                     |
| Employee Security Training               | KnowBe4 Security Awareness Training                                                                                   | 180 users        | \$20       | \$3,600     | N/A                                              | N/A                                         | Annual subscription for phishing simulations and training |
| Open-Source Software                     | Wazuh (SIEM), Snort (IDS/IPS), Zabbix (Monitoring), oTicket (Ticketing), FreeRADIUS (Authentication), Modsecurity WAF | N/A              | \$0        | \$0         | Deployed on existing servers                     | N/A                                         | Reduces software licensing costs                          |
| Total Costs                              |                                                                                                                       |                  |            | \$43,349.98 |                                                  |                                             |                                                           |



# Physical Organization & Access Controls



Note: This is a simplified illustration of devices used. For the complete network diagram, please refer to slide "Network Infrastructure".



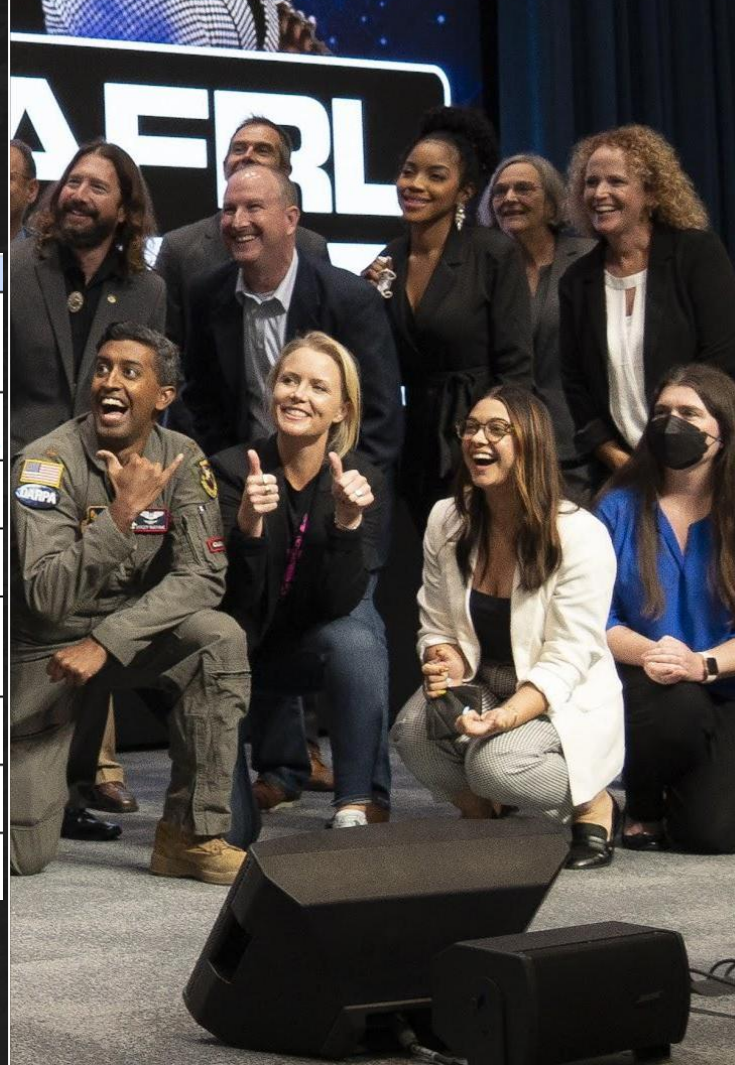
# Network Monitoring & Software

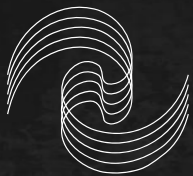
| Software                 | Hosted On            | Connected To                                                                                                                                    | Purpose                                                                                                                 |
|--------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Wazuh (SIEM)             | Private FTP Server   | <ul style="list-style-type: none"><li>- Core Switch</li><li>- All servers (AD, Web, Mail, DNS, FTP)</li><li>- Access switches</li></ul>         | Collects logs and telemetry from all endpoints and servers for correlation + anomaly alerts                             |
| Snort (IDS/IPS)          | Multi-Homed Firewall | <ul style="list-style-type: none"><li>- Border Router</li><li>- Core Switch</li><li>- DMZ</li><li>- VLAN interfaces (mirror/span)</li></ul>     | Real-time packet inspection and threat detection at perimeter and DMZ                                                   |
| Zabbix (Monitoring)      | Private DNS Server   | <ul style="list-style-type: none"><li>- DHCP Server</li><li>- FTP Server</li><li>- Core Switch</li><li>- All Access Switches</li></ul>          | Monitors performance/availability of network devices and servers                                                        |
| osTicket (Helpdesk)      | DHCP Server          | <ul style="list-style-type: none"><li>- VLAN 10 only (Privileged Helpdesk users)</li><li>- Core Switch</li></ul>                                | Web-based ticketing system accessible only to IT/helpdesk staff in VLAN 10                                              |
| FreeRADIUS (802.1X Auth) | Mail Server (DMZ)    | <ul style="list-style-type: none"><li>- Core Switch</li><li>- All Access Switches</li><li>- Wireless APs (if added)</li><li>- VLAN 10</li></ul> | Authenticates users via 802.1X before network access is granted                                                         |
| ModSecurity (WAF)        | Web Server (DMZ)     | <ul style="list-style-type: none"><li>- Border Router</li><li>- NGFW (Palo Alto PA-220)</li><li>- Switch 10</li><li>- Public Internet</li></ul> | Filters and blocks HTTP/S traffic using OWASP Top 10 rules; protects the web application from code injection, XSS, etc. |

# Expertise

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| Role / Expertise Area                        | Responsibility / Tasks                                                                                                                                                                    |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network Engineer (L2/L3)                     | <ul style="list-style-type: none"><li>- Design and configure VLANs across switches</li><li>- Implement inter-VLAN routing and trunking</li><li>- Apply ACLs for traffic control</li></ul> |
| Firewall & Security Appliance Specialist     | <ul style="list-style-type: none"><li>- Deploy and configure next-generation firewalls (NGFW)</li><li>- Define firewall zones, NAT rules, DMZ segmentation, and IDS/IPS</li></ul>         |
| Wireless Network Engineer                    | <ul style="list-style-type: none"><li>- Deploy enterprise wireless solution (WPA3-Enterprise, RADIUS)</li><li>- Eliminate rogue APs and optimize signal coverage</li></ul>                |
| Network Access Control (NAC) Engineer        | <ul style="list-style-type: none"><li>- Implement 802.1X port-based access control</li><li>- Integrate NAC with RADIUS and Active Directory/LDAP</li></ul>                                |
| System Administrator (Windows/Linux)         | <ul style="list-style-type: none"><li>- Patch and harden all servers</li><li>- Manage AD, DNS, DHCP scopes, and enforce Group Policies</li><li>- Decommission unsupported OSes</li></ul>  |
| Intrusion Detection Specialist               | <ul style="list-style-type: none"><li>- Set up and tune Suricata or equivalent IDS/IPS</li><li>- Configure alerting thresholds and log correlation</li></ul>                              |
| Monitoring & Logging Engineer                | <ul style="list-style-type: none"><li>- Deploy and maintain Zabbix, Graylog, or similar tools for SNMP, Syslog, and device health monitoring</li></ul>                                    |
| Cable Management / Infrastructure Technician | <ul style="list-style-type: none"><li>- Implement structured cabling and physical port mapping</li><li>- Set up rack enclosures and UPS</li></ul>                                         |





# Thanks



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# Appendix & Links

- [Firewalls Rules, Addressing, Costs, Expertise & Policies](#)